

Assorted Fun Classic Problems

UTSA Problem Solving Club

Which is Larger?

Without using a calculator, determine which number is larger:

$$2^{100} \quad \text{or} \quad 3^{75}.$$

Source: *MindYourDecisions*

100 Lockers

There are 100 closed lockers in a hallway and 100 students.

- Student 1 opens every locker.
- Student 2 closes every second locker (2, 4, 6, ...).
- Student 3 changes the state (opens if closed, closes if open) of every third locker (3, 6, 9, ...).
- This continues until Student 100 changes the state of the 100th locker.

Question: Which lockers are open at the end? Why?

Source: *Numberphile*

The Golden Root

Evaluate the following expression:

$$x = \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}$$

Question: What famous mathematical constant is this value equal to?

Source: *Tipping Point / Ramanujan*

The Three Semicircles

Three semicircles are placed along a line. Two smaller semicircles are tangent to each other and both are internally tangent to a larger semicircle. A vertical line segment is drawn from the point of tangency of the two small semicircles to the edge of the large semicircle.

Question: If the length of this vertical segment is h , find the area of the region inside the large semicircle but outside the two smaller ones (the “Arbelos” region) in terms of h .

Source: *MindYourDecisions*

Power Tower

Find the last digit of the number:

$$7^{7^7}$$

Hint: Look for a pattern in the last digits of $7^1, 7^2, 7^3, \dots$ and use modular arithmetic on the exponent.

Inscribed Square

Consider a right triangle with side lengths 3, 4, and 5. A square is inscribed in this triangle such that one side of the square lies on the hypotenuse (the side of length 5).

Question: What is the side length of this square?

Source: *3Blue1Brown (inspired)*

Sums of Consecutive Integers

In how many ways can the number 2026 be written as a sum of two or more consecutive positive integers?

Hint: If the sum is $n + (n + 1) + \dots + (n + k) = S$, use the formula for an arithmetic series and look for odd factors of $2S$.

Langley’s Adventitious Angles

Consider the isosceles triangle ABC with $\angle A = 20^\circ$ and $\angle B = \angle C = 80^\circ$. Points D and E are placed on sides AC and AB respectively, such that $\angle DBC = 60^\circ$ and $\angle ECB = 50^\circ$.

Question: Find the measure of $\angle BDE$.

Hint: This deceptively simple-looking problem stumped many professional mathematicians when first published. Direct angle-chasing alone won’t work—try an auxiliary construction.

Source: *E. M. Langley, Math. Gazette, 1922*

